

Divvy Bike-Share Case Study

Data Analysis Life Cycle: From Source SQL to Tableau Storytelling

Phase 1: Ask

The primary business objective is to analyze historical Divvy bike-share trip data over a full trailing 12-month period to identify distinct behavioral differences between casual riders and annual members. By unlocking these behavioral trends, the marketing team can design a data-driven conversion strategy targeted at converting casual riders into long-term subscribers.

Key Strategic Question:

How do annual members and casual riders use Divvy bikes differently, and what marketing triggers will incentivize a casual rider to purchase an annual membership?

Phase 2: Prepare

Data sourcing involved 12 distinct monthly CSV files extracted from zipped raw transactional archives. These files were compiled, normalized, and uploaded via a sequential staging pipeline into a MySQL relational database ecosystem for secure storage and structured querying.

Query 2.1: Master Dataset Row Audit

Executed to cross-examine total loaded database records against source CSV tallies:

```
SELECT COUNT (*) AS total_rides FROM bikes
```

Metric	Rows Returned
total_rides	5,848,703

Phase 3: Process

Before finalizing operational insights, data verification and cleaning protocols were implemented to isolate anomalies. This processing step addressed systemic false starts (trips lasting less than 60 seconds) and database processing errors resulting in negative trip durations (where the trip end timestamp precedes the start timestamp), ensuring an unskewed dataset for downstream visualization.

Phase 4: Analyze

Core Exploratory Data Analysis (EDA) was performed via SQL server aggregates to extract behavioral baselines across user segments.

Query 4.1: Rider Segment Proportions

Calculates the relative distribution of volume across user types:

```
SELECT member_casual, COUNT (*) AS total_rides,

        (COUNT (*) * 100.0 / (SELECT COUNT(*) FROM bikes),2) AS percentage

FROM bikes
GROUP BY member_casual;
```

member_casual	total_rides	percentage
casual	2,100,265	35.91%
member	3,748,438	64.09%

Query 4.2: Trip Duration Metric by Segment

Determines the average trip time in minutes via timestamp intervals:

```
SELECT member_casual, ROUND(AVG(TIMESTAMPDIFF(MINUTE, started_at, ended_at)), 2) AS
avg_duration_minutes
FROM divvy_trips GROUP BY member_casual;
```

member_casual	avg_duration_minutes
member	11.98
casual	21.77

Query 4.3: Weekly Usage and Seasonality Profile

Tracks behavioral differences aggregated across the days of the week:

```
SELECT member_casual, DAYNAME(started_at) AS day_of_week, COUNT(*) AS total_rides FROM
divvy_trips GROUP BY member_casual, day_of_week ORDER BY member_casual, FIELD(day_of_week,
```

```
'Sunday' 'Monday' 'Tuesday' 'Wednesday' 'Thursday' 'Friday'
'Saturday');
```

Day of Week	Member Rides	Casual Rides
Sunday	414,395	360,308
Monday	527,257	245,259
Tuesday	592,543	234,846
Wednesday	594,444	236,518
Thursday	601,521	267,111
Friday	548,705	323,399
Saturday	469,573	432,824

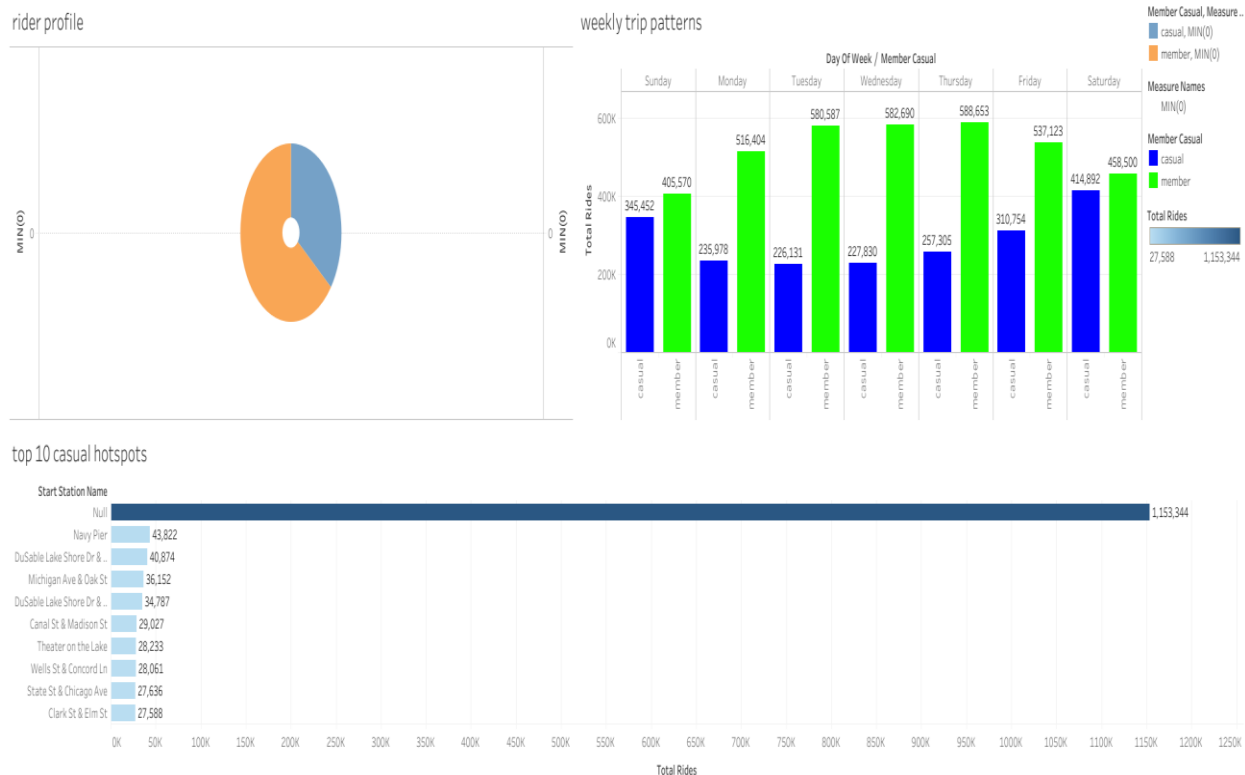
Query 4.4: Geographical Hotspot Analysis (Casual Riders)

Top 10 historical starting docking stations utilized exclusively by casual riders to maximize marketing conversion efficiency:

Rank	Start Station Name	Total Casual Rides
1	Navy Pier	38,364
2	DuSable Lake Shore Dr & Monroe St	32,263
3	Michigan Ave & Oak St	22,887
4	DuSable Lake Shore Dr & North Blvd	19,888
5	Millennium Park	18,821
6	Shedd Aquarium	16,861
7	Theater on the Lake	16,272
8	Dusable Harbor	15,338
9	Streeter Dr & Grand Ave	14,981

Rank	Start Station Name	Total Casual Rides
10	Michigan Ave & 8th St	11,055

Phase 5: Share



Phase 6: Act (Strategic Recommendations)

- **Message Shift (Highlights Savings):** Pivot marketing from work commutes to weekend leisure. Showing casual riders (who average **21.77-minute trips**) that a membership is cheaper than single-ride fees turns a hobby into an obvious financial win.
- **Targeted Ads (Eliminates Waste):** Concentrate physical signs and promo teams strictly at top casual hubs. Focusing the budget on hotspots like **Navy Pier, DuSable Lake Shore Dr, and Millennium Park** guarantees we hit the target audience without wasting ad spend.
- **Smart Alerts (Captures Peak Intent):** Trigger a single-click membership discount via the app the exact moment a rider locks their bike at a Top 10 station. This catches users at peak satisfaction, removing friction and driving immediate upgrades.